Bluefin tuna (*Thunnus thynnus*)

Summary Scores

- **Sensitivity:** Low
- **Exposure:** High
- **Vulnerability:** Low

Sensitivity			
Sensitivity Attributes	**Score**	**LH DQ**	**Tally**
Adult Mobility	1.08	3	
Complexity in Reproductive Strategy	2.08	2.8	
Dispersal of Early Life Stages	1.56	3	
Egg and Larval Survival and Settlement Requirements	2.56	2.8	
Historical Variability in Temperature	1.52	3	
Juvenile Habitat Requirements	1.12	3	
Other Stressors	1.28	2.7	
Population Growth Rate	2.4	2.8	
Prey Specificity	1.2	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.12	2.5	
Spawning Cycle	1.96	2.8	
Stock Size/Status	1.68	2.6	
Total Sensitivity | Certainty	**Low (1) | 0.67**		

Exposure			
Exposure Factors	**Mean**	**SD**	**Tally**
Sea Surface Temperature	2.65	1.28	
Sea Surface Salinity	2.94	0.94	
Surface pH	3.74	0.37	
Mixed layer depth (delta rho = 0.03)	1.09	0.38	
Net primary production	1.27	0.45	
Total Exposure	**High (2.93)**	**0.53**	

Vulnerability			
Vulnerability	**Mean**	**SD**	**Tally**
Total Vulnerability	**Low (2.93)**	**0.53**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Bluefin tuna. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Moderate (Loughran et al. 2025)	Low	Decrease
Exposure	High (Loughran et al. 2025)	High	None
Vulnerability	Moderate (Loughran et al. 2025)	Low	Decrease
Directional Effect	Neutral (Loughran et al. 2025)	Neutral	None

Stocks

Bluefin tuna is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

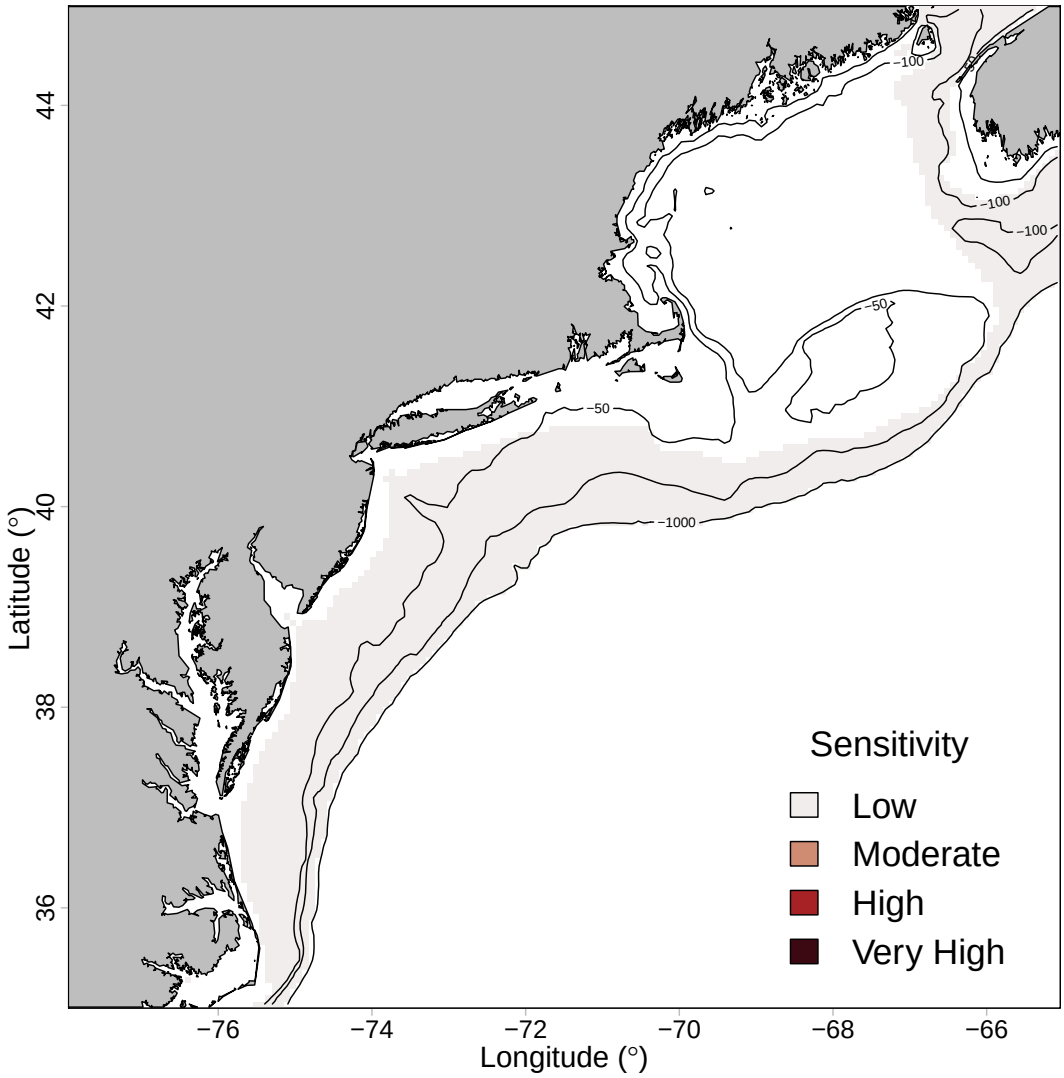


Figure 2: Bluefin tuna Sensitivity on the Northeast Shelf

Bluefin Tuna received a **Low** sensitivity score, with a certainty of 67% (Figure 1). The only attribute that received a score greater than 2.5 was Egg and Larval Survival and Settlement Requirements (2.6) (Figure 1). Early larval stages are susceptible to starvation and older larvae are susceptible to predation (Shropshire et al. 2022). Larvae also depend on ocean currents for transport to nursery habitats.

Bluefin Tuna were not included in the 2016 Climate Vulnerability Assessment for the Northeast Shelf, but were included in the recent Highly Migratory Species Climate Vulnerability Assessment (Loughran et al. 2025). In this analysis, Bluefin Tuna received a Moderate sensitivity score across their range (Loughran et al. 2025). This score was driven by population growth rates, reproductive cycles, and early life history requirements, similar to our analysis.

Exposure:

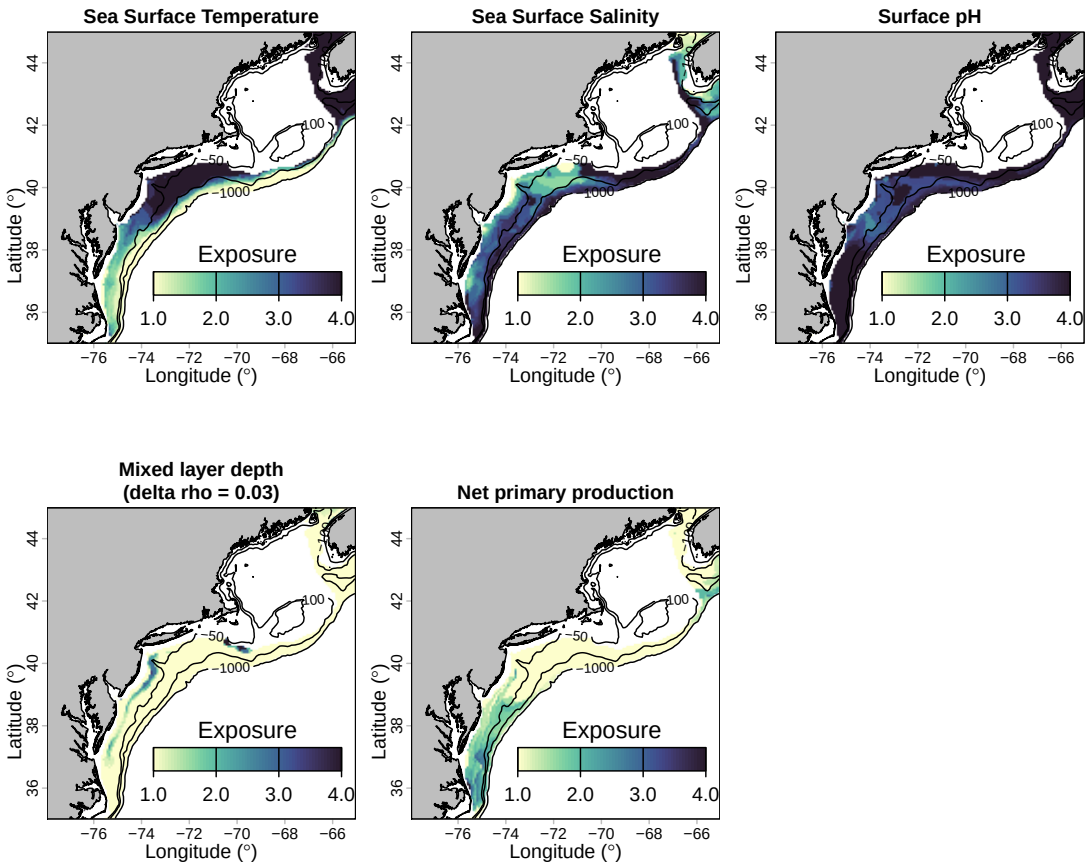


Figure 3: Maps of each Exposure Factor for Bluefin tuna. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure for Bluefin Tuna was **High** (2.93 +/- 0.53) across its distribution along the Northeast Shelf (Figure 1). This was driven by high sea surface temperature, salinity, and pH exposure (Figure 3). Exposure was relatively uniform across the species' range, with pockets of Moderate exposure off the coasts of the DelMarVa Peninsula, and pockets of High Exposure along the edge of the Gulf of Maine (Figure 4).

Bluefin Tuna were not assessed in the previous Climate Vulnerability Assessment for the Northeast Shelf, but were recently assessed by the Highly Migratory Species group (Loughran et al. 2025). In that assessment, which included their entire distribution throughout the Northwest Atlantic, they received a High Exposure score (Loughran et al. 2025). This score was driven by pH, surface oxygen, and surface temperature (Loughran et al. 2025).

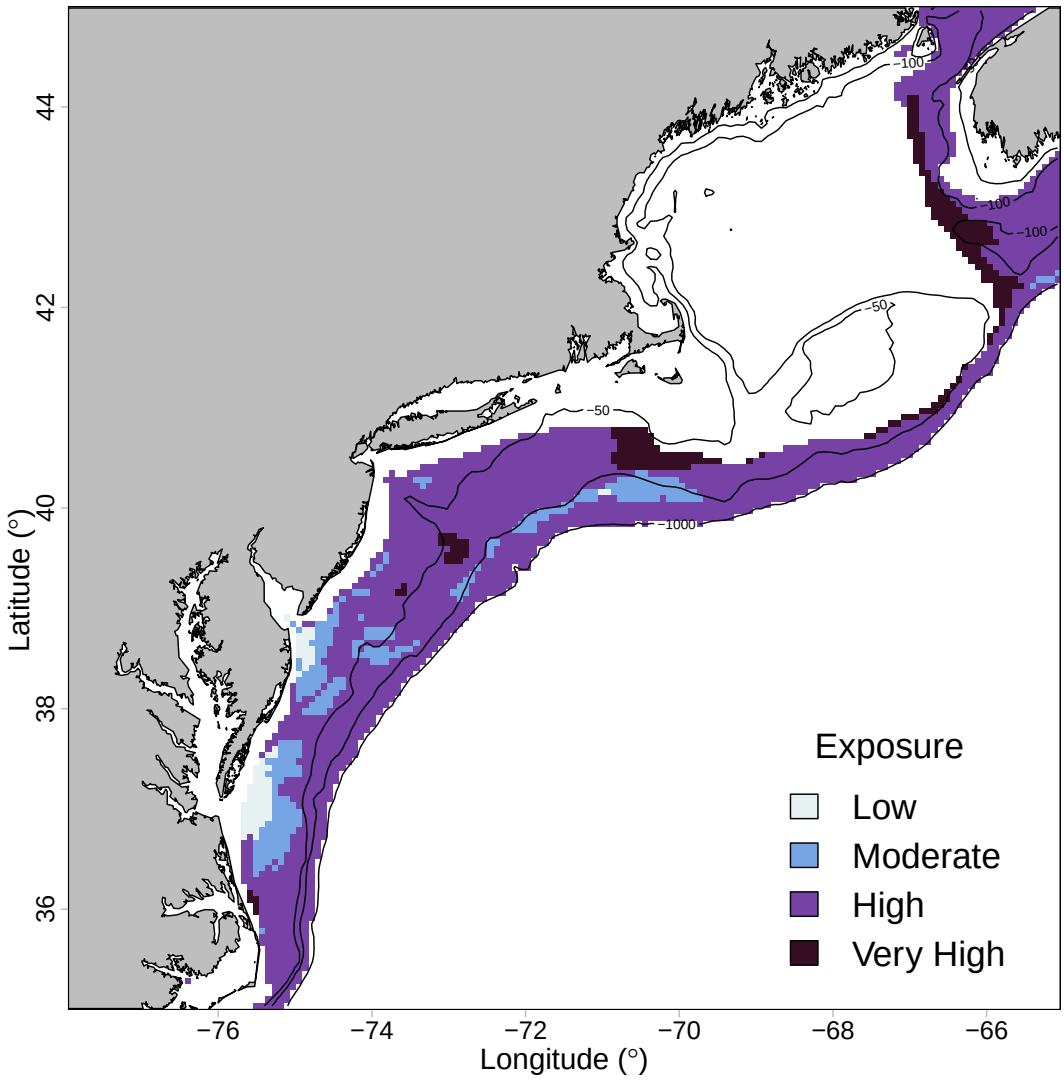


Figure 4: Total Exposure on the Northeast Shelf for Bluefin tuna. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

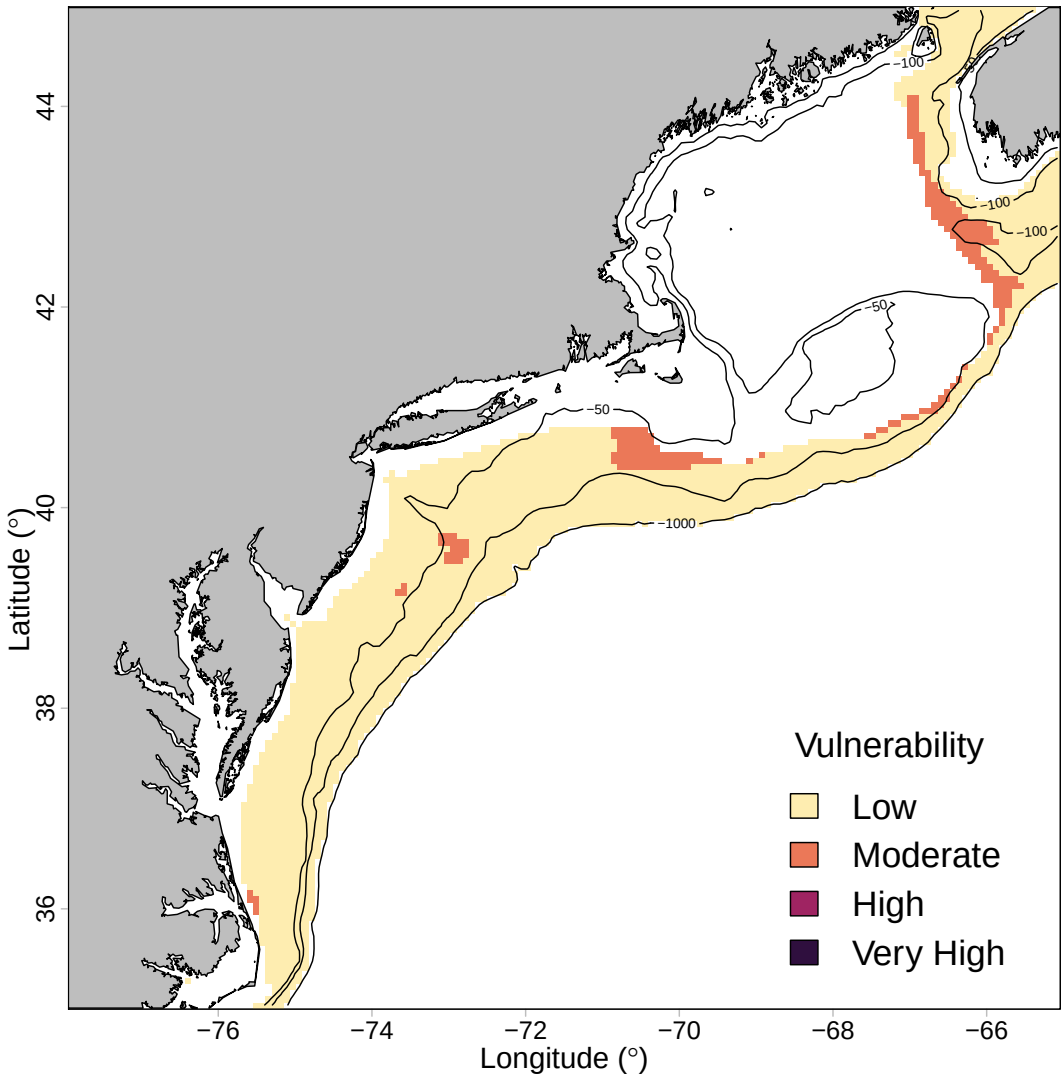


Figure 5: Vulnerability on the Northeast Shelf for Bluefin tuna. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

The mean Vulnerability score for Bluefin Tuna was **Low** (2.93 +/- 0.53) due to the Low sensitivity score (Figure 1). Vulnerability was Low throughout the species' distribution along the Northeast Shelf ((**fin-vuln-bft?**); Figure 1).

Bluefin Tuna received a Moderate Vulnerability score in the recent highly migratory species climate vulnerability assessment (Loughran et al. 2025). They were not assessed in the previous climate vulnerability assessment for the Northeast Shelf (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

98% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

The bluefin tuna received a mean model confidence score of 0.5 (+/- 0.5), indicating no confidence in the model. While experts stated that broad trends matched expectations, the low sample size, highly migratory behaviors of bluefin tuna, and a lack of predictions in known habitats drove the score down. The ensemble model AUC was 0.98 indicating that the model had excellent predictive skill.

Additional Metrics

Distribution Shifts:

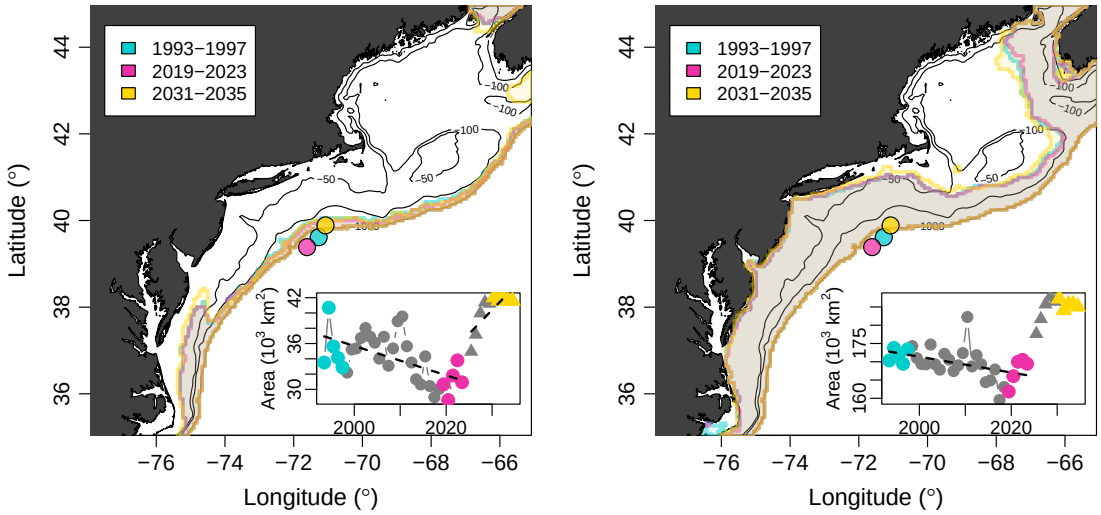


Figure 6: Distribution shifts on the Northeast Shelf for Bluefin tuna. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Between 1993 and 2023 there were significant declines in bluefin tuna habitat area on the Northeast Shelf (Figure 6). The weighted centroid shifted 39 km southwest during this time.

Core habitat area for bluefin tuna is predicted to expand significantly due to increased habitat area off the coast of Maryland and Virginia and as north as Nova Scotia (Figure 6). The weighted centroid is predicted to shift a further 73 km to the northeast between 2025 and 2035.

Distribution Vulnerability was not calculated in the recent Highly Migratory Species Climate Vulnerability Assessment (Loughran et al. 2025).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	35,842	172,272	8.54	9.54
2019-2023	38.7 km SW	31,332 (-4,510)	167,717 (-4,554)	8.85 (+0.31)	9.54 (0)
2031-2035	72.5 km NE	41,818 (+10,485)	185,684 (+17,967)	9.04 (+0.19)	9.62 (+0.08)

Directional Effect:

Bluefin tuna were estimated to be **neutrally** impacted by climate change. Expert certainty was 96%. Bluefin distributions have been projected to move poleward into the northeast Atlantic. Habitat losses are expected to be great (60% loss) in the Gulf of America, but smaller (20%) in the Northwest Atlantic (Braun et al. 2023). Spawning may shift to early spring due to increasing temperatures (Muhling et al. 2011). This is similar to the results from the recent Highly Migratory Species Climate Vulnerability Assessment (Loughran et al. 2025).

Climate Effects on Abundance and Distribution

Projected surface temperatures in the Gulf of America may alter the spatial and temporal extent of Atlantic bluefin tuna spawning. Model results from (Muhling et al. 2011) indicate decreased larval bluefin tuna occurrence in late spring in the Gulf by 39-61% by 2050 and 93-96% by 2100, with early spring spawning becoming more prevalent. Erauskin-Extramiana et al. (2026) also modeled decreased habitat suitability for spawning in the Gulf of America. They predict a poleward shift in Atlantic bluefin tuna distribution and greater spatial overlap with prey in higher latitudes. Nursery habitats in the Gulf of America are impacted by oil spills. Upwards of 5% of predicted spawning habitat within the U.S. Exclusive Economic Zone (EEZ) was impacted by the Deepwater Horizon oil spill (Hazen et al. 2016).

Life History Synopsis

Western Atlantic bluefin tuna are a highly migratory species observed from Newfoundland to the Gulf of America; they occupy temperate surface waters but can dive to depths of 1000m (Fromentin 2006). Spawning occurs between spring and summer throughout western North Atlantic spawning grounds, namely the Gulf of America (March-June) and Slope Sea (June-September). Spawning in the Slope Sea occurs one to two months later than the Gulf of America due to differences in sea surface temperatures (Baglin 1982). Bluefin tuna eggs hatch after 2 days and larvae remain pelagic for 3-4 weeks (Shropshire et al. 2022). Larval bluefin tuna are found near spawning grounds and dispersed via wind and the margins of ocean currents. Edges of ocean currents and eddies provide access to warmer sea surface temperatures and nutrient-rich waters that support larval foraging of zooplankton prey while avoiding predation (Shropshire et al. 2022). Favorable habitat for bluefin larvae in the Gulf of America was determined to be

moderately warm waters (23.5-28.5°C) in the eddies of the Loop Current, and outside of cooler, higher chlorophyll continental shelf waters (Teo, Boustany, and Block 2007). Along the coast of the southeastern United States, larval stages may benefit from nearshore meanders of the Gulf Stream that can produce upwelling of nutrient rich water along the continental shelf edge (Shropshire et al. 2022). Larvae initially feed on zooplankton and progress to a piscivorous diet at a relatively small size (Llopiz and Hobday 2015). Starvation is the largest cumulative source of mortality in early larval stages, and predation on older larvae is a main source of mortality in later stages (McGowan and Richards 1989). Juvenile habitats are located over the continental shelf in the summer and further offshore in the winter. Juvenile life stages follow similar seasonal movements compared to adult life stages, though are generally more restricted to coastal areas (National Marine Fisheries Service (NMFS) 2006). Juvenile and adult bluefin tuna are opportunistic, generalist predators feeding mainly on zooplanktivorous fishes and crustaceans. Predation of larger fish continues with increasing body size, with juveniles moving from sand lance to mackerel and herring as adults (Logan et al. 2011) (Logan, Golet, and Lutcavage 2015). Adult bluefin tuna migrate seasonally from winter to early summer spawning grounds in the Gulf of America, Straits of Florida, and Slope Sea to summer and fall foraging grounds throughout the Gulf Stream, South and Mid-Atlantic Bight, Gulf of Maine, Nova Scotia Shelf, and east of the Flemish Cap (Block et al. 2001). Age at 50% maturity estimates for Atlantic bluefin tuna vary between 5 to 12 years, with a maximum age of over 30 years (Standing Committee on Research and Statistics (SCRS) 2021). NOAA Fisheries, with recommendations from the International Commission for the Conservation of Atlantic Tunas (ICCAT), manages the Atlantic bluefin tuna fishery in the United States through the Atlantic Highly Migratory Species Management Division. As of 2023, the stock is being managed ICCAT Recommendation 21-10. Population numbers are unknown but the stock is not subject to overfishing (International Commission for the Conservation of Atlantic Tunas (ICCAT) 2022).

Supplemental Scores

Exposure Timeseries

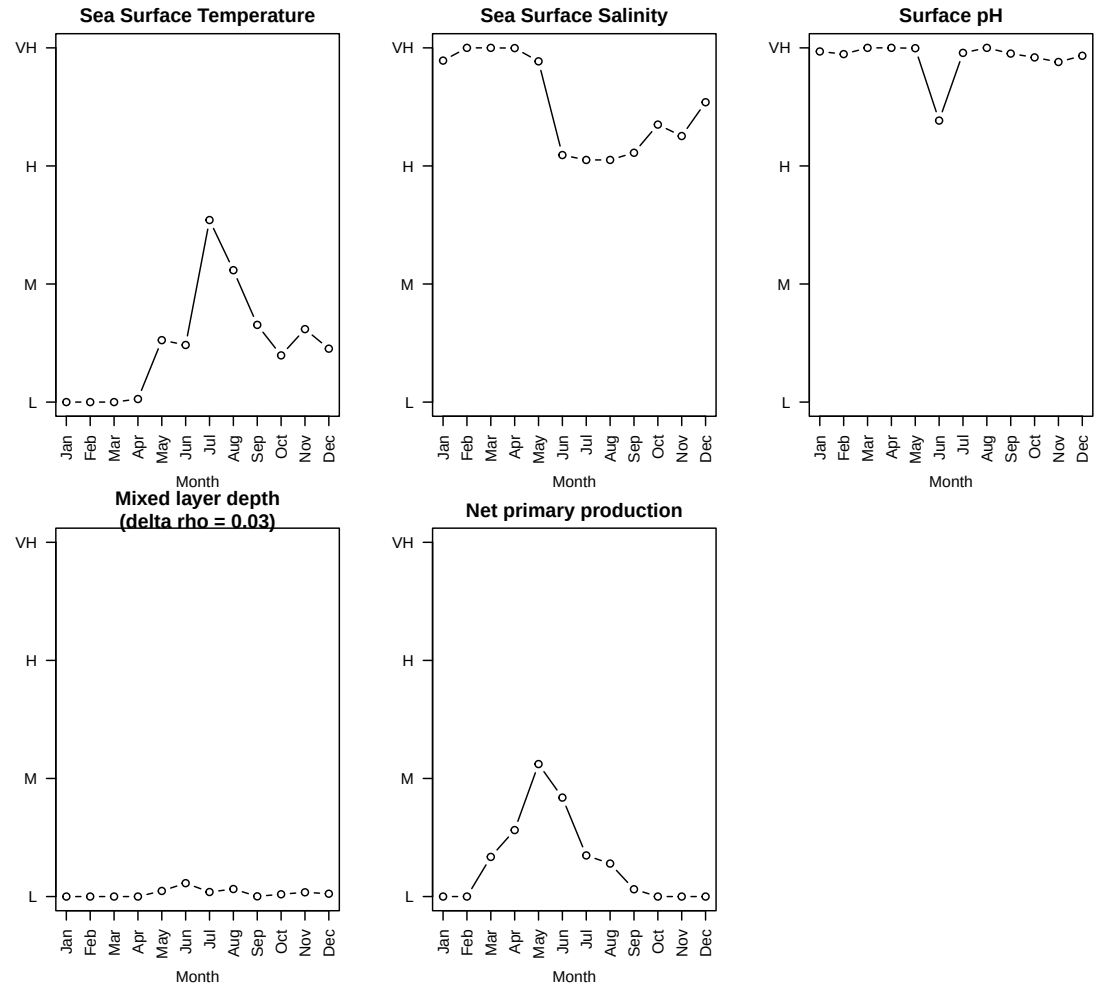


Figure 7: Monthly Timeseries of each Exposure Factor for Bluefin tuna. Exposure was calculated between 2014-2023 and 2025-2035

Exposure remained High across all months. There was no month-to-month variability (Figure 8). There was some variability in individual factors, but the overall Exposure score did not change (Figure 7).

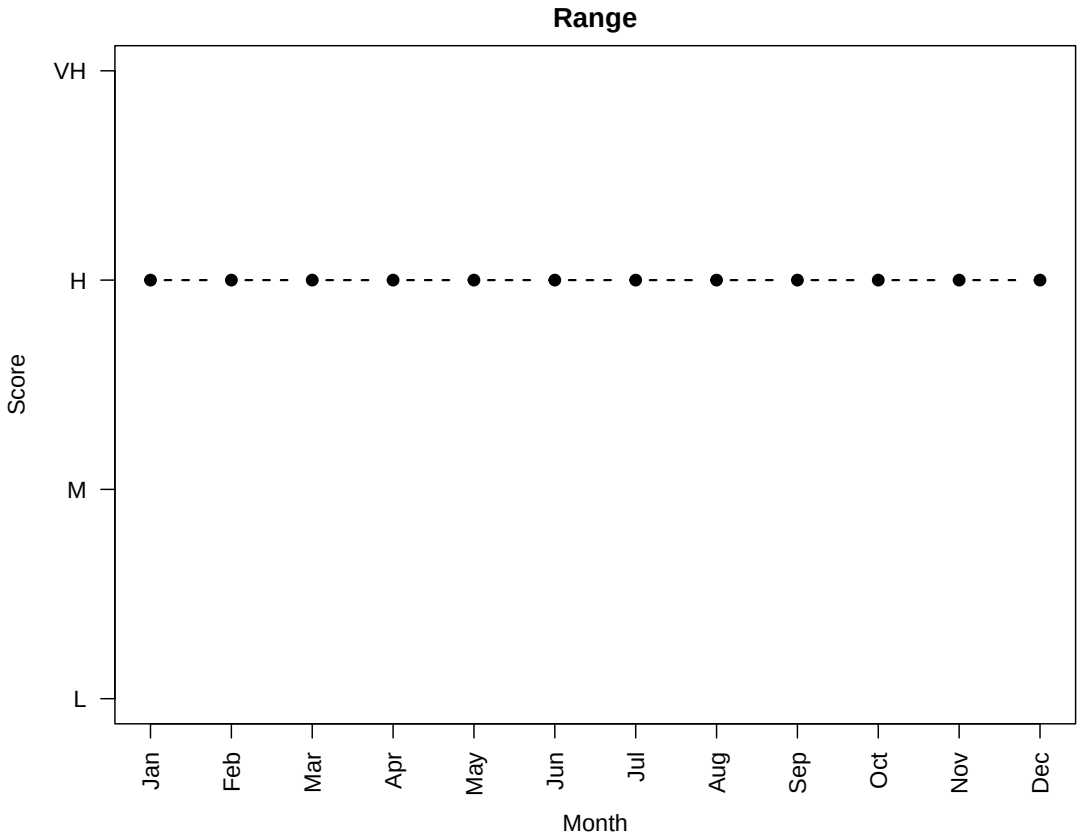


Figure 8: Timeseries of Total Exposure for Bluefin tuna

Exposure due to Significant Factors

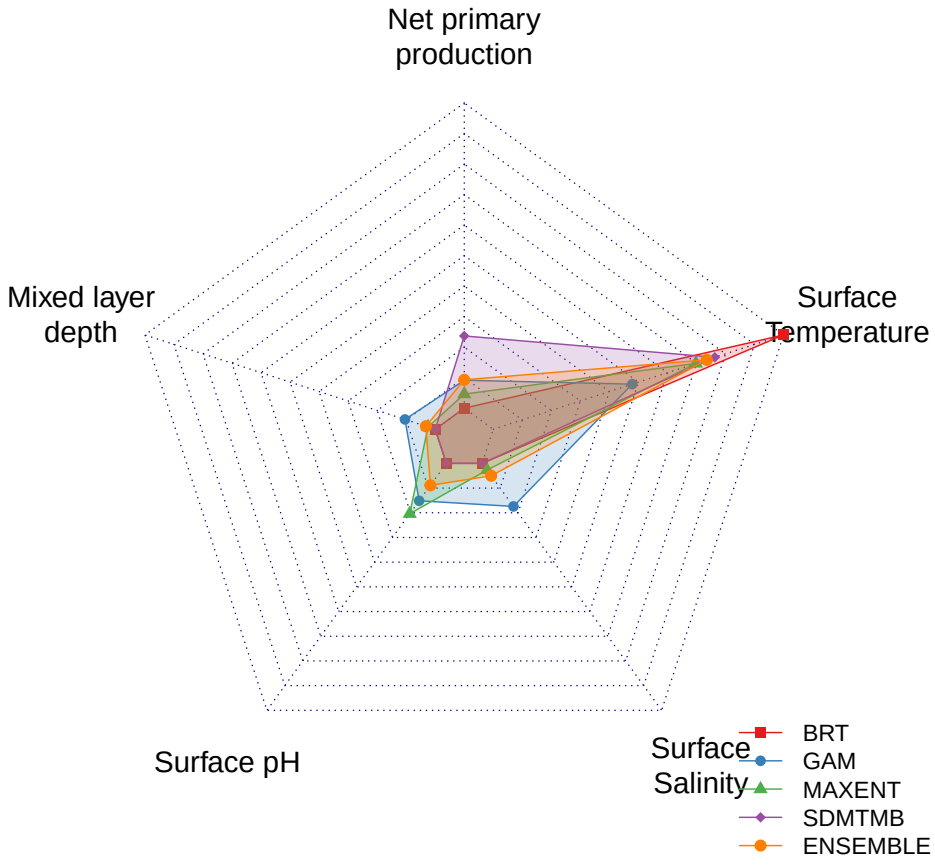


Figure 9: Dynamic Factors driving Bluefin tuna distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Sea Surface Temperature was the only major driver of Bluefin Tuna distribution along the Northeast Shelf (Figure 9). As a result, Exposure was Low when calculated with only important factors due to the limited number of important factors (Figure 10; Figure 11).

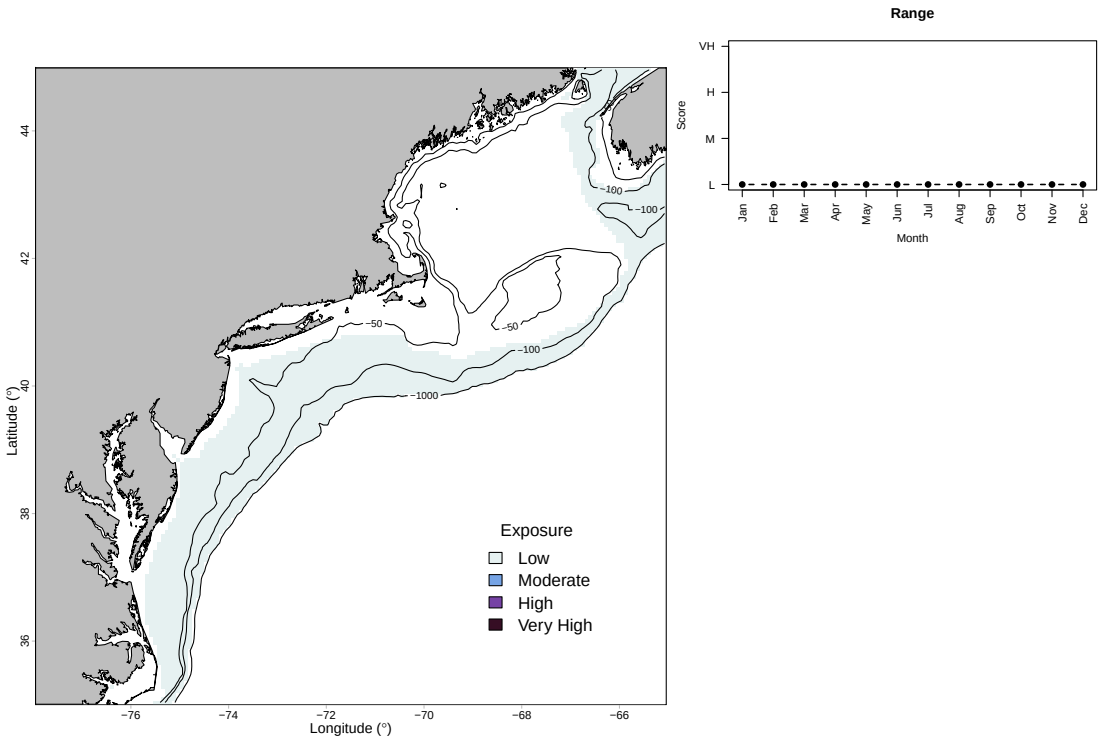
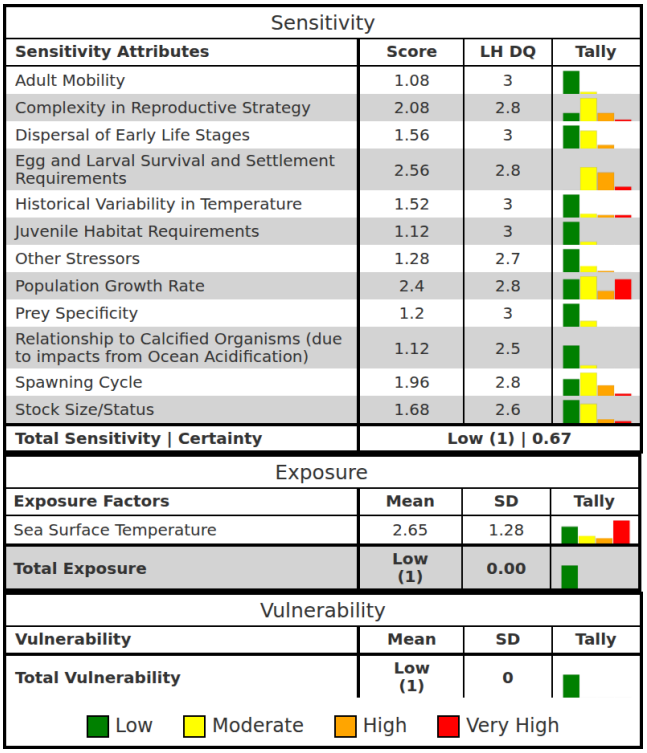


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Bluefin tuna distributions



Sensitivity			
Sensitivity Attributes	**Score**	**LH DQ**	**Tally**
Adult Mobility	1.08	3	
Complexity in Reproductive Strategy	2.08	2.8	
Dispersal of Early Life Stages	1.56	3	
Egg and Larval Survival and Settlement Requirements	2.56	2.8	
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Spawning Cycle	1.96	2.8	
Stock Size/Status	1.68	2.6	
Total Sensitivity | Certainty	**Low (1) | 0.67**		

Exposure			
Exposure Factors	**Mean**	**SD**	**Tally**
Sea Surface Temperature	2.65	1.28	
Total Exposure	**Low (1)**	**0.00**	

Vulnerability			
Vulnerability	**Mean**	**SD**	**Tally**
Total Vulnerability	**Low (1)**	**0**	

Low Moderate High Very High

Figure 11: As in Figure 1 for Bluefin tuna, but with only significant Exposure factors and the resulting Vulnerability Scores.

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